MetaGraphSci – Milestone 1

Graph Transformer Self-Supervised Contrastive Neighbor Adaptation
for Scientific Document Classification

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Problem Identification and Motivation

Scientific document classification is the task of assigning the subject category of a research document based on its content as well as its context within the larger scholarly literature. It is a significant task with applications in digital libraries, semantic search, massive document indexing, and recommendation systems. It is also a challenging task due to the lack of sufficient supervision data, as the classification of a document with respect to a fine-grained classification system requires domain knowledge, and only a small fraction of the documents in the real world are annotated.

A scientific document is a rich medium that offers three sources of information: the *textual signal*, which is the semantic content of the document; the *bibliographic signal*, which includes the publication venue, publisher, authorship, and publication year; and the *relational signal*, which includes the network of citations between documents. A good classifier should not consider the document as just a sequence of words but as a rich object with its context within the larger scholarly literature.

Existing methods only partially exploit this information. Text encoders such as SciBERT (Beltagy et al., 2019) and citation-informed encoders such as SPECTER (Cohan et al., 2020) provide strong semantic representations, but downstream classification is still often driven primarily by text. Graph-based models capture citation dependencies, yet their performance may deteriorate in low-label settings because structural aggregation can blur discriminative semantics, especially when the graph is noisy or weakly homophilous (Rampasek et al., 2022; Wu et al., 2022). Similarly, citation-aware contrastive learning methods improve representation quality through structural supervision, but they usually define similarity only through graph proximity and do not fully exploit metadata as an informative source of relational evidence (Ostendorff et al., 2022; Taechoyotin and Acuna, 2024).

These constraints give rise to the need to develop a semi-supervised pipeline called **MetaGraphSci** that combines text, metadata, and citation context. The main hypothesis is that scientific document classification can be improved by combining semantic encoders with metadata interactions and graph contextualization. The proposed pipeline has three principal components: a scientific text encoder that is pretrained, a metadata interaction encoder, and a citation context transformer. The training is working in two-stage: a metadata-informed contrastive pretraining with neighborhood awareness, followed by adaptive pseudo-label fine-tuning in scenarios with limited labeled data.

Literature Review

Scientific Document Representation from Text

One of the primary methodologies that this field focuses on is the use of pre-trained models that are trained on scientific text data. An example of this would be the use of the SciBERT model (Beltagy et al., 2019), which makes use of the vocabulary and corpus that are specific to the domain of scientific discourse. However, the SciBERT model remains at the top of the list for general document representations in the scientific domain. Furthermore, the SPECTER model (Cohan et al., 2020) builds on this approach by taking into consideration the citation information during the pre-training phase, where it learns to put papers with similar citation patterns close together in the embedding space, which remains a major improvement over the existing state of the art in citation-aware document representations.

SciNCL (Ostendorff et al., 2022) continued to improve upon the effectiveness of the contrastive learning approach by sampling from the K-hop citation neighborhood of each paper instead of relying on co-citation relationships. This work is of particular relevance to the proposed model as it highlights the importance of neighborhood-aware contrastive learning as opposed to batch-aware contrastive learning. Also recent works propose that scientific documents should not simply be treated as text sequences. CoSAEmb (Singh and Singh, 2024) proposed section-aware contrastive representations of scientific documents, while MISTI (Taechoyotin and Acuna, 2024) showed the effectiveness of incorporating metadata into the contrastive learning approach. This indicates that more effective representations of scientific documents can be achieved by making use of structure beyond the title and abstract of a paper.

Graph Neural Networks and Graph Transformers

Citation graphs provide a natural foundation for modeling relationships between scholarly documents, and graph neural networks have therefore become a standard approach for node-level tasks in this domain. But, traditional message-passing architectures exhibit notable limitations. In particular, repeated local aggregation tends to homogenize node representations, leading to the well-known problem of over-smoothing as network depth increases. This effect is especially detrimental in low-label settings, where preserving semantic discriminability is critical.

GraphGPS (Rampasek et al., 2022) addresses this issue by combining local message passing with global attention, resulting in a hybrid architecture that balances expressiveness with scalability. Similarly, NodeFormer (Wu et al., 2022) demonstrates that transformer-based reasoning on graphs can be made efficient through learned structure induction and attention approximations. More broadly, the development of graph transformers with explicit structural priors has been explored in Yun et al. (2019) and further generalized by Dwivedi and Bresson (2022). These works highlight the importance of incorporating structural encodings, such as topological and spectral features, to enhance transformer performance on graph-structured data. This perspective directly informs the design of the citation-context encoder in MetaGraphSci.

Semi-Supervised Learning and Pseudo-Labeling

Because offered labels are scarce in scientific corpora, semi-supervised learning is central to the problem. Weakly supervised methods such as that of Meng et al. (2020) showed that useful classifiers can be bootstrapped even when explicit annotations are limited, highlighting the value of unlabeled data when handled carefully.

More recent work has focused on pseudo-label selection. Yang et al. (2025) proposed class-adaptive pseudo-label thresholds, showing that fixed global thresholds bias selection toward dominant classes and exacerbate class imbalance. Similarly, Chuang et al. (2024) showed that monitoring pseudo-labeling status across epochs can reduce confirmation bias and improve sample selection. These works are directly relevant to the second training stage of MetaGraphSci, where reliable pseudo-label generation is required under imbalance and uncertainty.

Fine-Grained Scholarly Classification Benchmarks

For fine-grained scientific classification, it is necessary that the data reflects the semantic diversity of the information contained in the data, as well as the inherent class imbalance that is present in the data, which reflects the real world. FoRC4CL (Ahmad et al., 2024a) is particularly relevant in this regard, as it provides a manually curated data set of NLP papers, classified according to a fine-grained field of research taxonomy. The analyses that follow (Ahmad et al., 2024b; Dampa, 2025; Francis et al., 2025) reveal that hierarchical interdependencies between classes, as well as class imbalance, are key concerns that persist with regard to the actual classification task, making the data set useful not only for evaluating the accuracy of the model, but also its ability to retain discriminative ability under real-world, intricately structured conditions.

Best Existing Datasets and Solutions

Datasets

In this study, FoRC4CL (Ahmad et al., 2024a) is seen as a prominent evaluation resource for this specific problem, considering that it is designed to classify NLP articles at a detailed level. The hierarchical structure and class imbalance properties of this specific benchmark make it particularly useful as a resource to evaluate the robustness of a system under practical circumstances. Finally, with regards to citation network benchmarks such as **Cora**, **PubMed**, and **ogbn-arxiv** it is noted that they remain applicable within the context of graph node classification problems. Even though they are not as specific as FoRC4CL, they remain appropriate controlled environments to compare various structural modeling techniques with each other, as well as to test their scalability.

Best Existing Solutions

Among text-and citation-aware encoders, SciNCL (Ostendorff et al., 2022), as one of the state-of-the-art models in this space, is particularly prominent in this setting due to the combination of scientific language modeling with neighborhood-aware contrastive supervision. The development context of SciNCL is particularly relevant given the more nuanced consideration of meaningful positive context as opposed to citation-aware context.

Among graph-based models, the balance of local relational information and global transformer attention in GraphGPS (Rampasek et al., 2022) is particularly robust, as is the scalable transformer variant of NodeFormer (Wu et al., 2022). These models are particularly relevant as structural baselines to MetaGraphSci. In the space of semi-supervised optimization, the pseudo-labeling strategy based on class adaptivity proposed by Yang et al. (2025) is particularly pertinent to the second training stage of this proposed idea. While not originally motivated by citation graph document classification, the underlying principle is particularly relevant to this low-label scholarly context.

Comparative Analysis and Proposed Improvements

Although the above methods have substantially advanced scientific document representation learning, several limitations remain open.

First, **metadata is still underutilized**. Methods such as SPECTER and SciNCL define positive relations primarily through citation structure. However, in real scholarly corpora two documents may be strongly related without citing one another directly, for example because they share venue, publication period, or authorship patterns. This motivates the introduction of an explicit metadata affinity function $\mu(i, j)$, which refines the notion of document similarity beyond the citation graph alone.

Second, **graph transformers rarely model bibliographic compatibility explicitly**. Existing graph-transformer architectures generally rely on node features and structural encodings, but do not directly incorporate metadata as a bias in attention. In scientific corpora, however, bibliographic compatibility often acts as an informative relational prior. Injecting such compatibility into attention is therefore a natural extension for scholarly document modeling.

Third, **pseudo-labeling is usually treated as an instance-wise decision problem**. Standard pseudo-label mechanisms rely only on confidence estimates and do not explicitly account for the fact that graph neighbors may occupy semantically related regions of the label space. This suggests that graph-aware calibration and selection are especially important in citation-based corpora.

Limitations of Existing Methods	Improvements in MetaGraphSci
Text-only or weakly graph-aware models do not jointly exploit semantic, structural, and bibliographic evidence.	Unified multimodal modeling over text, citation context, and metadata.
Citation-aware contrastive learning usually defines similarity only from graph proximity.	Metadata-aware similarity through an explicit affinity function $\mu(i, j)$.
Graph-transformer attention is typically independent of bibliographic compatibility.	Metadata-conditioned attention bias integrated into the citation-context encoder.
Contrastive learning may treat graph-neighbor documents as negatives, creating false-negative pressure.	Neighborhood-aware masking of the negative pool during contrastive pretraining.
Pseudo-label selection often relies on fixed or weakly adaptive thresholds.	Class-adaptive pseudo-labeling with confidence calibration and dynamic thresholds.

Table 1: Comparison between existing limitations and the design goals of MetaGraphSci.

Identified Alternatives

Some of these alternatives were considered during the design of MetaGraphSci.

For the text encoder, a citation-aware model such as SPECTER (Cohan et al., 2020) could be used as the primary backbone. However, SciBERT remains a strong and flexible choice, particularly when citation structure is modeled explicitly during contextualization rather than fully embedded in pretraining.

For the graph encoder, GraphGPS (Rampasek et al., 2022) and NodeFormer (Wu et al., 2022) provide competitive solutions with different trade-offs in scalability and expressiveness.

In contrast, our approach focuses on directly integrating bibliographic compatibility into graph reasoning, instead of treating structural and metadata signals separately.

Regarding the training objective, purely citation-based contrastive learning, as in SciNCL (Ostendorff et al., 2022), represents a strong baseline. We extend this idea by incorporating metadata-based compatibility, enabling a more comprehensive notion of similarity aligned with the multimodal nature of scientific documents.

Finally, while standard pseudo-labeling methods (Meng et al., 2020; Chuang et al., 2024; Yang et al., 2025) can be applied independently, we adopt a different perspective: pseudo-labeling should be tightly coupled with the learned multimodal representation, leveraging text, graph structure, and metadata jointly. Overall, MetaGraphSci is based on the premise that scientific documents are not isolated text units, but structured entities embedded in citation networks and enriched by bibliographic information. Effective classification therefore requires integrating these signals in a unified manner.

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